The Collatz map inside its family: machine-verified structure of $f(x) = ax + c$

GPU-scale certified dynamics, a universal-cycle law characterizing $a = 2^k - 1$, and the arithmetic $2^H - a^k$ that organizes every cycle

J. Naude, with derivations, engines, and analyses produced in human–AI collaboration (Claude, Anthropic; see the repository commit trailers). Computational artifacts, datasets, and machine proofs accompany this paper: the CUDA engine `collatz.cu`, the Lean 4 developments `CollatzTheory.lean` and `CollatzCerts.lean`, the exact-arithmetic cross-checker `analyze.py`, and the raw sweep data in `results/`.

Working paper, revision of 2026-07-03. The Collatz conjecture is ***open****; nothing here proves or disproves it, and §10 argues that no route of this kind can. The contribution is a machine-verified structural account of the two-parameter family containing it: every theorem is checked by the Lean 4 kernel, every finite claim is certified, and every empirical law is measured on $\approx 3.76 \times 10^{10}$ GPU-computed orbits.*

Abstract

For odd a, c we study the generalized Collatz map

$$T_{a,c}(n) = \begin{cases} a\,n + c & n \text{ odd,} \\ n/2 & n \text{ even,} \end{cases}$$

the classical conjecture being the case $(a, c) = (3, 1)$. We combine a CUDA sweep of 38 parameter pairs — every odd start below 2^{32} for the $3x + c$ family and below 2^{30} otherwise, $\approx 3.76 \times 10^{10}$ orbits in about 7 seconds on one RTX 5090 — with a Lean 4 development in which (i) the structural theorems are proved for *all* parameters with kernel-only proofs and no extra axioms, and (ii) the sweep's finite claims are exported as 127 machine-checked certificates (89 cycle certificates, axiom-free by kernel evaluation; 38 range-classification certificates via `native_decide`). A second engine (`universal.cu`) verifies the sharpened universal-cycle laws of §4.1 on a further $\approx 4.1 \times 10^{11}$ instances with zero violations, and `verify_universal.py` re-derives them in exact arithmetic up to 350-digit parameters.

The verified structure is: **(1)** if a or c is even, every odd start diverges monotonically, so the family is dynamically interesting only for a, c odd; **(2)** for odd m, $T_{a,mc}(mn) = m\,T_{a,c}(n)$, so cycle sets scale, and a common odd prime factor of a and c traps all orbits in $p\mathbb{Z}$ — reducing the classification to $\gcd(a, c) = 1$; **(3)** if $a = 2^k - 1$ then for *every* odd c the point c itself is periodic, $T^{k+1}(c) = c$: the famous loop $1 \to 4 \to 2 \to 1$ is the $k = 2$ instance of a one-line theorem. Sharpened (§4.1): the accelerated map F satisfies $F_{a,c}(c) = \mathrm{odd}(a+1) \cdot c$, so the universal fixed point at c *characterizes* $a = 2^k - 1$; the fixed points of any (a, c) are exactly $x = c/d$ for divisors $d \mid c$ with $a + d$ a power of two; and the master identity $F_{a,c}(cy) = c\,F_{a,1}(y)$ transports every $ax + 1$ cycle into every $ax + c$ — a sweep of all odd $a < 2^{24}$ finds exactly 25 multipliers with a universal cycle through c: the 24 Mersenne numbers (period 1) and $a = 5$ (period 2, the family $\{c, 3c\}$). Empirically, the halving count after an odd step has mean 2 (measured 2.0000–2.0009 on escape-dominated ensembles), giving per-odd-step drift $\delta(a) = \ln a - 2 \ln 2$, which the sweep reproduces to four decimals for every $a \geq 5$ together with the implied mean escape times. Consequently the family splits into a contracting regime — all twelve $3x + c$ systems tested converge for 100.0000% of $\approx 2.1 \times 10^9$ odd starts each — and an expanding regime, where convergence density is $\leq 0.36\%$ and often exactly the cycle basins. Every one of the 89 cycles found satisfies the exact identity $n_0(2^H - a^k) = c\,W$ with $W = \sum_i a^{k-1-i} 2^{h_0 + \cdots + h_{i-1}}$, and the inventories are organized by the divisors of $D = 2^H - a^k$: one-step cycles exist iff $(2^h - a) \mid c$, which simultaneously explains the universal cycles ($D = 1$), the isolated fixed points of $(9, 7)$, $(11, 5)$, $(13, 3)$ ($2^4 - a = c$), the seven-cycle multiplet of $3x + 13$ ($2^8 - 3^5 = 13$), and the emptiness of the $(9, 1)$, $(11, 1)$, $(13, 1)$ inventories. We state the two conjectures this evidence supports and record why they — and by Conway-type undecidability, the general family — remain beyond computational and elementary methods.

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1. Introduction

1.1 The conjecture and its family

The Collatz (3x+1) conjecture asserts that iterating $T_{3,1}$ from any positive integer reaches the cycle $1 \to 4 \to 2 \to 1$. It has been verified for all $n < 2^{71}$ (Barina 2020, 2025) and holds for "almost all" orbits in the strong sense of Tao (2019), yet remains open. We embed it in the two-parameter family $T_{a,c}$ above, asking what *can* be settled about the family as a whole — by proof, by machine-checked certificate, and by exhaustive measurement — and where exactly the residual hardness lives.

Two classical facts calibrate the ambition. First, the base case is open despite half a century of effort; the strongest general results are density statements (Terras 1976; Everett 1977; Korec 1994; Tao 2019). Second, Conway (1972) showed that generalized Collatz-type functions simulate arbitrary computation, and Kurtz–Simon (2007) made the generalized problem Π^0_2-complete: **no uniform decision procedure for the family exists.** "Solving the general case" can therefore only mean: prove every structural theorem that is provable, certify every finite claim, and reduce the remainder to sharply-stated conjectures with quantitative evidence. That is the shape of this paper.

1.2 Contribution

1. **Machine-verified general theorems** (§4): parity reduction, scaling conjugacy, prime absorption, and a universal-cycle theorem for $a = 2^k - 1$ — sharpened in §4.1 to an iff, with a complete one-step-cycle law and a transport principle for cycles — each proved in Lean 4.31 with kernel-only proofs, no Mathlib, no `native_decide`, quantified over all parameters.
2. **A certified computational pipeline** (§5): a two-pass CUDA engine (cycle inventory by Brent detection; mass classification of every odd start) whose output is cross-checked in exact big-integer arithmetic and *re-proved* as 127 Lean certificates (§5.3). The GPU finds; the proof assistant certifies.
3. **Quantitative laws** (§6–§8): the drift dichotomy $\delta(a) = \ln a - 2\ln 2$ measured to four decimals; the cycle equation verified exactly on all discovered cycles; the signature/divisor mechanism $D = 2^H - a^k$ that predicts which (a, c) have rich, sparse, or empty cycle inventories.
4. **An honest synthesis** (§9–§10): the two open conjectures the data supports, and a precise account of why this route — any computational route — cannot close them.

2. Related work

Terras (1976) and Everett (1977) proved almost every n has finite stopping time; Korec (1994) lowered the a.e. dip to n^θ, $\theta = \ln 3/\ln 4 \approx 0.7925$. Tao (2019) proved almost all Collatz orbits attain almost bounded values (logarithmic density). Computational verification reached 2^{68} (Barina 2020) and 2^{71} (Barina 2025), which also feeds cycle bounds: Steiner (1977) excluded 1-circuits; Simons–de Weger (2005) excluded m-cycles for $m \le 68$; Hercher (2023) for $m \le 91$, so any nontrivial $(3,1)$-cycle has length exceeding $\approx 10^{11}$. Lagarias (1985, 2010) surveys the field; Lagarias (1990) identified integer cycles of $3x + d$ with rational cycles of $3x + 1$. Belaga–Mignotte (1998, 2000) catalogued the primitive cycles of $3x + d$ for $d < 20000$ and conjectured every such orbit is eventually periodic. Crandall (1978) formulated the $qx + 1$ problem and conjectured divergent orbits exist for $q \ge 5$. Conway (1972; FRACTRAN 1987) and Kurtz–Simon (2007) supply the undecidability ceiling. The cycle identity of §7 is classical (Böhm–Sontacchi 1978). A community effort to formalize the Collatz literature in Lean is underway at ccchallenge.org; the present development is independent but uses the same proof assistant.

3. Notation

For odd n, one *odd step* $n \mapsto an + c$ is followed by $h = \nu_2(an + c) \ge 1$ halvings (a, c odd makes $an + c$ even); the *accelerated map* is

$$F(n) = \frac{a\,n + c}{2^{\nu_2(an+c)}},$$

an odd-to-odd map. A cycle with k odd members $n_0, \ldots, n_{k-1}$ and halving exponents $h_0, \ldots, h_{k-1}$ has *signature* (k, H), $H = \sum h_i$, and length $k + H$ as a T-cycle. Computations classify odd iterates against a *window* τ: $\tau = 2^{62}$ for $a \le 3$ and $\tau = \lfloor (2^{64} - 1024)/a \rfloor$ otherwise, chosen so $an + c$ never overflows 64-bit arithmetic. An orbit *escapes* when an odd iterate exceeds τ; even peaks may lawfully exceed τ without escaping. The one negative constant swept, $c = -1$, is the classical $3x - 1$ system (equivalently, Collatz on the negative integers).

4. Machine-verified structural theorems

All results in this section are proved in `CollatzTheory.lean` for **all** parameters (not merely tested values), type-checked by the Lean 4.31 kernel. `#print axioms` reports only the standard `propext`, `Classical.choice`, `Quot.sound`; there is no `sorry`, no Mathlib, and no compiled-code trust. Proof sketches follow the formal proofs.

Theorem 1 (parity reduction). *Let n be odd and positive. If a is odd and c even ($c > 0$), or a is even ($a \ge 2$) and c odd, then $T_{a,c}(n)$ is odd and $T_{a,c}(n) > n$; consequently the orbit is strictly increasing forever:*

$$T^{j}_{a,c}(n) \;\ge\; n + j \qquad (j \ge 0).$$

(Lean: `step_growth_evenC`, `step_growth_evenA`, `orbit_diverges_evenC`.)

Proof. an has the parity of $a \cdot n$; in both hypothesis regimes $an + c$ is odd, and $an + c > n$ since $a \ge 1, c > 0$ resp. $a \ge 2$. The even branch is never taken, so growth iterates. ☐

Thus the family is dynamically nontrivial **only for** a, c **both odd**, which fixes the sweep grid.

Theorem 2 (scaling conjugacy). *For odd m and all a, c, n, j:*

$$T_{a,\,mc}(m\,n) = m\,T_{a,c}(n), \qquad T^{j}_{a,\,mc}(m\,n) = m\,T^{j}_{a,c}(n).$$

(Lean: `scaling_step`, `scaling_orbit`.) In particular $m \cdot \mathrm{Cycles}(a, c) \subseteq \mathrm{Cycles}(a, mc)$.

Proof. mn has the parity of n; on the odd branch $a(mn) + mc = m(an + c)$, on the even branch $(mn)/2 = m(n/2)$. Induct. ☐

Theorem 3 (prime absorption). *If an odd p divides both a and c, then $p \mid T_{a,c}(n)$ for every odd n, and $p \mid n$ implies $p \mid T_{a,c}(n)$ unconditionally; hence after one odd step every orbit lies in $p\mathbb{Z}$ forever. (Lean: `absorb_entry`, `absorb_step`, `absorb_orbit`.)*

With Theorem 2 this reduces the classification to $\gcd(a, c) = 1$: e.g. every $(9, 3)$-orbit is $3\times$ a $(9, 1)$-orbit after its first odd step.

Theorem 4 (universal trivial cycle). *Let $a + 1 = 2^k$ (i.e. $a = 3, 7, 15, 31, \ldots$). Then for* every *odd c,*

$$T^{k+1}_{a,c}(c) = c,$$

realized as $c \;\mapsto\; (a+1)c = 2^k c \;\mapsto\; \cdots \;\mapsto\; c$. (Lean: `halve_iter`, `universal_cycle`, instantiated as `trivial_cycle_3/7/15` with c universally quantified.)

Proof. c odd gives $T(c) = ac + c = (a+1)c = 2^k c$; then k even steps halve $2^k c$ back to c. ☐

The Collatz loop $\{1, 4, 2\}$ is the $k = 2$, $c = 1$ instance; the theorem asserts its analogue in *every* $3x + c$, $7x + c$, $15x + c$ system simultaneously, and the sweep observes it in all 21 such variants (§6).

4.1 The universal cycle, sharpened: formula, converse, classification, transport

Sketching the sweep of Theorem 4 exposed a stronger mechanism, which one identity carries entirely. Write $\mathrm{odd}(n) = n/2^{\nu_2(n)}$ for the odd part, so the accelerated map of §3 is $F_{a,c}(x) = \mathrm{odd}(ax + c)$. For **odd** c and *all* a, y:

$$\boxed{F_{a,c}(c\,y) \;=\; c \cdot F_{a,1}(y)} \qquad \text{in particular} \qquad F_{a,c}(c) \;=\; \frac{a+1}{2^{\nu_2(a+1)}}\,c \;=\; \mathrm{odd}(a+1) \cdot c.$$

The proof is the observation itself: $a(cy) + c = c\,(ay + 1)$, and an odd factor passes through the odd part — the point c *factors out of its own orbit*. Everything below is this identity read three ways, each Lean-proved for all parameters (kernel-only, as in §4).

Theorem 4b (master formula and Mersenne rigidity). *For every odd c and every a: $F_{a,c}(c) = \mathrm{odd}(a+1)\cdot c$. Consequently*

$$F_{a,c}(c) = c \iff \mathrm{odd}(a+1) = 1 \iff a = 2^k - 1,$$

so the sufficient condition of Theorem 4 is also necessary — and already for a single odd c, hence simultaneously for all of them. (Lean: `F_formula`, `universal_fixed_iff`, `F_universal_cycle`; T-form `generalized_universal_step`: if $a + 1 = 2^k m$, m odd, then $T^{k+1}(c) = mc$.)

Theorem 4c (one-step cycle law). *For odd c: x is a fixed point of $F_{a,c}$ iff $d\,x = c$ and $a + d = 2^k$ for some divisor d of c — fixed points correspond exactly to the divisors $d \mid c$ with $a + d$ a power of two, via $x = c/d$ — and every such x is a genuine T-cycle of length $k + 1$. (Lean: `fix_classification`, `F_fix_realizes_T`.)*

The two k=1 rows of Table 2 are the extreme divisors, now theorems rather than observations: $d = 1$ forces a Mersenne and gives the universal fixed point $x = c$; $d = c$ forces $a + c \in 2^{\mathbb{N}}$ and gives the isolated fixed point $x = 1$ of $(9, 7), (11, 5), (13, 3)$. The law also *proves* the eight empty inventories of §6 have no one-step cycles: none of those (a, c) admits a divisor $d \mid c$ with $a + d$ a power of two.

Theorem 4d (orbit transport: universal cycle families). *For odd c and all a, y, n:*

$$F^n_{a,c}(c\,y) = c \cdot F^n_{a,1}(y), \qquad \text{and} \qquad F^n_{a,c}(cy) = cy \iff F^n_{a,1}(y) = y.$$

The $ax + 1$ system embeds in every $ax + c$ system, rescaled by c, and periodicity transports in both directions. (Lean: `F_scaling` — proved in the general form $F_{a,\,de}(du) = d\,F_{a,e}(u)$ for odd d — `F_orbit_scaling`, `F_periodic_transport`.)

Theorem 4d reframes Theorem 4: the universal cycle at c is the transported trivial fixed point of $ax + 1$, and **every** cycle of $ax + 1$ is a *universal cycle family* of the whole column $\{(a, c) : c \text{ odd}\}$. Two consequences the original theorem could not see:

- $a = 5$ is not Mersenne, yet every $5x + c$ system has the two-cycle $\{c, 3c\}$ — the transport of the $5x + 1$ cycle $\{1, 3\}$ (Lean: `five_universal_two_cycle`);
- every $181x + c$ system has the two-cycle $\{27c, 611c\}$ — the transport of the classical $181x + 1$ cycle $\{27, 611\}$, a cycle *not through* 1 (Lean: `oneEightyOne_universal_two_cycle`).

It also poses a sharp finite question: *which multipliers have a universal cycle through the seed c itself?* By Theorem 4d that set is exactly $\{a : 1 \text{ is periodic under } F_{a,1}\}$, which §6.1 sweeps to 2^{24}.

5. Computational methodology

5.1 CUDA engine

`collatz.cu` runs two kernels per variant on an NVIDIA RTX 5090 (Blackwell, `sm_120`, driver CUDA 13.2; compiled as `compute_90` PTX and JIT-compiled by the driver, since the installed `nvcc` 12.0 predates the architecture):

- **Pass A — inventory.** Brent cycle detection on the accelerated map F from every odd start $< 2^{22}$ (fuel 2^{14} accelerated steps, window τ). Detected cycles are canonicalized by their odd minimum; (k, H), the cycle maximum, and members are recomputed on the host.
- **Pass B — classification.** Every odd start $< N$ ($N = 2^{32}$ for the $3x + c$ family, 2^{30} otherwise) is iterated (fuel 8192) until its odd value hits an inventory minimum (basin attribution), exceeds τ (escape), or fuel exhausts. Aggregates per variant: per-cycle basin counts, escape count, total odd steps, total halvings, maximum excursion; per-thread tallies are reduced through shared memory. Grid total: 38 variants, $\approx 3.76 \times 10^{10}$ orbits, ≈ 7 s of GPU time.

A second engine, `universal.cu`, verifies the §4.1 theorems at scale (RTX 5090, run of 2026-07-03; raw output `results/universal.jsonl`):

- **S1 — rigidity grid.** All 2.75×10^{11} odd pairs (a, c), $a, c < 2^{20}$: the master formula $F_{a,c}(c) = \mathrm{odd}(a+1)\,c$ and the Mersenne iff hold with **zero** violations; the fixed-point count is exactly $20 \cdot 2^{19}$ (20 Mersenne a below 2^{20} times 2^{19} odd c) — 1.0 s.
- **S2 — fixed-point census.** All 1.37×10^{11} triples ($a, c < 512$, $x < 2^{22}$): brute enumeration finds 5477 fixed points; the set coincides *exactly* with the divisor-law prediction of Theorem 4c, enumerated independently on the host — 0.5 s.
- **S3 — universal-family catalog.** Every odd $a < 2^{24}$: is 1 periodic under $F_{a,1}$? (§6.1.)
- **S4 — identity fuzz.** 2^{33} pseudorandom instances of $F_{a,\,de}(du) = d\,F_{a,e}(u)$ (d, e odd, a, u arbitrary): zero violations.

`verify_universal.py` then re-derives every S1–S4 claim in exact big-integer arithmetic (V1–V8): set-level equality of the census with the law, exact re-iteration of the whole catalog with the $n_0 = 1$ cycle equation, and stress instances of the formula, rigidity, transport, and one-step law at 80–350-digit parameters. All pass (`results/universal_verify.log`).

5.2 Exact-arithmetic cross-checks

`analyze.py` re-runs, in Python big-integer arithmetic (no windows, no overflow), every sampled orbit that left the u64 window or exhausted fuel. Fuel-outs: zero across all variants. **Every sampled window-escapee of the $a \le 3$ systems reconverges** — e.g. $n = 3{,}735{,}036{,}913$ under $3x + 15$ exits the 2^{62} window and returns to the cycle with minimum 57. The script further asserts the literature anchors ($3x + 1 \Rightarrow \{1\}$; $3x - 1 \Rightarrow \{1, 5, 17\}$; $5x + 1 \Rightarrow \{1, 13, 17\}$; $3x + 5 \supseteq \{1, 5, 19, 23\}$), the scaling identities of Theorem 2 on the data ($\mathrm{mins}(3, 15) = 3 \cdot \mathrm{mins}(3, 5)$ elementwise, etc.), and the cycle equation of §7 exactly on all 89 cycles.

5.3 Lean certificates

`analyze.py` exports the sweep's finite claims to `CollatzCerts.lean`:

- **89 cycle certificates** — for each discovered cycle, `iterN (T a c) len min = min`, proved by kernel `decide`: the Lean kernel itself evaluates the orbit. `#print axioms` reports **no axioms whatsoever** for these.
- **38 range certificates** — for each variant, *every* $n \in [1, 10^5]$ either reaches the certified inventory or exceeds τ within the fuel bound (`native_decide`, which additionally trusts Lean's compiler via `Lean.ofReduceBool` — the standard trade-off for large finite checks). Corollary: **inventory completeness** below the window — any missed cycle must have minimum $> 10^5$ (indeed $> 2^{22}$ by Pass A) or an odd element above τ.

All 127 certificates check (≈ 6 minutes; long cycles need `maxRecDepth` raised, as `decide` unfolds roughly ten elaborator frames per step).

6. Results

Table 1 condenses `results/summary.md` (full raw data: `results/raw.jsonl`). "conv." is the fraction of odd starts below N that reach the certified inventory inside the window; for $a = 3$ the window escapees were individually re-verified to reconverge (§5.2), making the convergence rows exact.

a	c	N	cycles (odd minima)	conv. %	drift meas.	drift pred.
1	1	2^{30}	$\{1\}$	100	−1.3863	−1.3863
3	−1	2^{32}	3: 1, 5, 17	100	−0.3045	−0.2877
3	1	2^{32}	1: $\{1\}$	100	−0.2845	−0.2877
3	3	2^{32}	1: $\{3\}$	100	−0.2843	−0.2877
3	5	2^{32}	6: $1, 5, 19, 23, 187, 347$	100	−0.2919	−0.2877
3	7	2^{32}	2: $5, 7$	100	−0.3059	−0.2877
3	9	2^{32}	1: $\{9\}$	100	−0.2841	−0.2877
3	11	2^{32}	3: $1, 11, 13$	100	−0.2994	−0.2877

a	c	N	cycles (odd minima)	conv. %	drift meas.	drift pred.
3	13	2^{32}	10: $1, 13, 131, 211, 227, 251, 259, 283, 287, 319$	100	-0.3013	-0.2877
3	15	2^{32}	6: $3, 15, 57, 69, 561, 1041$	100	-0.2921	-0.2877
3	17	2^{32}	3: $1, 17, 23$	100	-0.2996	-0.2877
3	19	2^{32}	2: $5, 19$	100	-0.3037	-0.2877
5	1	2^{30}	3: $1, 13, 17$	0.0851	$+0.2228$	$+0.2231$
5	3	2^{30}	7: $1, 3, 39, 43, 51, 53, 61$	0.0749	$+0.2233$	$+0.2231$
5	5	2^{30}	3: $5, 65, 85$	0.1489	$+0.2225$	$+0.2231$
5	7	2^{30}	6: $1, 7, 9, 57, 91, 119$	0.2877	$+0.2228$	$+0.2231$
5	9	2^{30}	10: $1, 3, 9, 29, 89, 117, 129, 153, 159, 183$	0.3594	$+0.2232$	$+0.2231$
5	11	2^{30}	5: $1, 11, 141, 143, 187$	0.1474	$+0.2228$	$+0.2231$
7	1	2^{30}	1: $\{1\}$	0.0001	$+0.5596$	$+0.5596$
7	3	2^{30}	1: $\{3\}$	0.0001	$+0.5596$	$+0.5596$
7	5	2^{30}	3: $3, 5, 27$	0.0012	$+0.5596$	$+0.5596$
7	7	2^{30}	1: $\{7\}$	0.0004	$+0.5596$	$+0.5596$
7	9	2^{30}	2: $1, 9$	0.0001	$+0.5596$	$+0.5596$
7	11	2^{30}	2: $11, 23$	0.0009	$+0.5596$	$+0.5596$
9	1	2^{30}	**none**	0.0000	$+0.8109$	$+0.8109$
9	3	2^{30}	**none**	0.0000	$+0.8109$	$+0.8109$
9	5	2^{30}	**none**	0.0000	$+0.8109$	$+0.8109$
9	7	2^{30}	1: $\{1\}$	~ 0	$+0.8109$	$+0.8109$
9	9	2^{30}	**none**	0.0000	$+0.8109$	$+0.8109$
9	11	2^{30}	**none**	0.0000	$+0.8109$	$+0.8109$
11	1	2^{30}	**none**	0.0000	$+1.0116$	$+1.0116$
11	3	2^{30}	**none**	0.0000	$+1.0116$	$+1.0116$
11	5	2^{30}	1: $\{1\}$	~ 0	$+1.0116$	$+1.0116$
13	1	2^{30}	**none**	0.0000	$+1.1787$	$+1.1787$
13	3	2^{30}	1: $\{1\}$	~ 0	$+1.1787$	$+1.1787$
15	1	2^{30}	1: $\{1\}$	~ 0	$+1.3218$	$+1.3218$
15	3	2^{30}	1: $\{3\}$	~ 0	$+1.3218$	$+1.3218$
15	5	2^{30}	1: $\{5\}$	~ 0	$+1.3218$	$+1.3218$

Table 1. The 38-variant sweep. Drift is per odd step: measured $\ln a - (\text{halvings}/\text{odd steps}) \ln 2$ vs. predicted $\delta(a) = \ln a - 2 \ln 2$ (§8).

Headlines: **(i)** every $a = 3$ variant converged for 100.0000% of its $\approx 2.1 \times 10^9$ odd starts (window escapes bignum-verified); **(ii)** every $a \geq 5$ variant escapes for essentially all starts, the convergent share being the union of tiny cycle basins; **(iii)** eight variants — $(9, 1), (9, 3), (9, 5), (9, 9), (9, 11), (11, 1), (11, 3), (13, 1)$ — have *empty* inventories: no positive cycle with minimum below 2^{22} (GPU) resp. certified below 10^5 (Lean) and odd elements below $\tau \approx 2^{61}$; even the orbit of 1 climbs beyond the window; **(iv)** the universal cycle of Theorem 4 appears in all 21 variants with $a \in \{3, 7, 15\}$, with signature $k = 1$, $H = k_2$ where $a + 1 = 2^{k_2}$ — and the catalog sweep of §6.1 shows the Mersennes and $a = 5$ are the *only* multipliers below 2^{24} whose universal family passes through the seed c itself; **(v)** $\text{Cycles}(3, 15) = 3 \cdot \text{Cycles}(3, 5)$ element-by-element with identical signatures — Theorem 2 live in data.

6.1 The universal-cycle catalog

Theorem 4d converts "which multipliers give a cycle through c in *every* $ax + c$?" into a one-parameter question: for which odd a is 1 periodic under the $ax + 1$ accelerated map? Pass S3 answers it for all 8,388,608 odd $a < 2^{24}$ (fuel 4096 odd steps, values windowed at $\approx 2^{64}/a$; **zero** fuel-outs, so every orbit either returned to 1 or left the window — no undecided cases):

verdict for the orbit of 1 under $F_{a,1}$	count	members
periodic, period 1	24	exactly $a = 2^k - 1$, $k = 1, \ldots, 24$ — forced by Theorem 4b
periodic, period 2	1	$a = 5$: $1 \to 3 \to 1$, transporting to $\{c, 3c\}$ in every $5x + c$
escaped the $2^{64}/a$ window	8,388,583	everything else

Table 1b. The universal-cycle catalog for $a < 2^{24}$. Every entry was re-iterated in exact big-integer arithmetic, satisfies the $n_0 = 1$ cycle equation $2^H - a^k = W$ (§7), and was re-verified as a live cycle of $F_{a,c}$ at 100-digit c (transport in action). Completeness caveat, stated honestly: an escaping orbit could in principle return to 1 from above the window; the sweep excludes periodicity only within the $2^{64}/a$, 4096-step bounds (drift $\delta(a) > 0$ makes a return implausible, but that is §8 heuristics, not proof.)

The period-2 row is itself an arithmetic law. A two-cycle through 1 must be $\{1,\ \mathrm{odd}(a{+}1)\}$, and it closes iff $a \cdot \mathrm{odd}(a+1) + 1$ is a power of two ($a = 5$: $5 \cdot 3 + 1 = 16$; Mersenne a degenerate with $\mathrm{odd}(a{+}1) = 1$). The sweep says no other $a < 2^{24}$ manages this — or any longer return — inside the window. Note the catalog does **not** exhaust universal families: $a = 181$'s orbit of 1 escapes, yet $\{27c, 611c\}$ is universal for the 181-column by transport of the cycle $\{27, 611\}$, which avoids 1. The catalog enumerates families through the seed c; families through $c\,y$, $y > 1$, correspond to the full cycle inventories of the $ax + 1$ systems (e.g. $5x + 1$'s $\{13, 33, 83\}$ and $\{17, 43, 27\}$ give $\{13c, 33c, 83c\}$ and $\{17c, 43c, 27c\}$ in every $5x + c$ — visible in Table 1 as the c-multiples populating the $a = 5$ rows).

7. The cycle equation organizes every inventory

Proposition 5 (cycle identity; classical, cf. Böhm–Sontacchi 1978). *Let $n_0 \to n_1 \to \cdots \to n_{k-1} \to n_0$ be a cycle of the accelerated map with halving exponents h_i, $H = \sum h_i$. Then, exactly,*

$$n_0 \left(2^H - a^k\right) \;=\; c\,W, \qquad W = \sum_{i=0}^{k-1} a^{\,k-1-i}\, 2^{\,h_0+\cdots+h_{i-1}}.$$

The identity was verified in exact arithmetic for all 89 discovered cycles (`analyze.py`), which simultaneously cross-checks the GPU's (k, H) bookkeeping. Its arithmetic content: writing $D = 2^H - a^k$ for the *signature denominator*, integer cycles with signature (k, H) exist iff D divides $c\,W$ for some admissible parity vector, so inventories are governed by the divisors of D — that is, by how well powers of 2 approximate powers of a. Observed instances:

mechanism	$D = 2^H - a^k$	consequence (observed)
$k = 1$, $2^h - a = 1$ ($a = 2^h - 1$)	$D = 1$	universal cycle at $n = c$, all 21 variants with $a \in \{3, 7, 15\}$ (Theorem 4; an **iff** by Theorem 4b)
$k = 1$, $2^h - a = c$	$D = c$	fixed point $n = 1$: exactly the systems $(9, 7)$, $(11, 5)$, $(13, 3)$ via $2^4 - a = c$ — the *only* $a \in \{9, 11, 13\}$ systems with any cycle (the $d = c$ case of Theorem 4c)
$k = 1$, $(2^h - a) \nmid c$ for all h	—	no one-step cycles — a theorem by 4c; combined with sparse higher signatures: the eight **empty** inventories of §6

mechanism	$D = 2^H - a^k$	consequence (observed)
$(k, H) = (5, 8)$, $a = 3$	$D = 2^8 - 3^5 = 13$	$c = 13$: a seven-cycle multiplet $211, 227, 251, 259, 283, 287, 319$, all of signature $(5, 8)$
$(k, H) = (3, 5)$, $a = 3$	$D = 2^5 - 3^3 = 5$	$c = 5$: the cycle pair $19, 23$ (and, scaled by 3: $57, 69$ in $c = 15$)
$(k, H) = (2, 4)$, $a = 3$	$D = 2^4 - 3^2 = 7$	$c = 7$: the cycle $\{5, 11\}$
$(k, H) = (3, 7)$, $a = 5$	$D = 2^7 - 5^3 = 3$	five signature-$(3, 7)$ cycles each for $(5, 3)$ and $(5, 9)$; the classical pair $\{13, 33, 83\}, \{17, 43, 27\}$ of $5x + 1$
$(k, H) = (2, 5)$, $a = 5$	$D = 2^5 - 5^2 = 7$	one such cycle in *every* $5x + c$ swept; two for $c = 7$

Table 2. Signature denominators explain rich, sparse, and empty inventories.

The deep $3x + 1$ cycle results are this mechanism run in reverse: Steiner, Simons–de Weger, and Hercher bound how close $2^H/3^k$ can approach 1 (continued fractions of $\log_2 3$, Baker-type bounds), forcing $|D|$ so large that small nontrivial cycles are impossible. In our data the same principle appears as: signatures cluster on the line $H/k \approx \log_2 a$, and inventories are nonempty exactly when the resulting D has the right divisors relative to c.

8. The drift law, measured

After an odd step, the exponent $h = \nu_2(an + c)$ of a "random" orbit behaves geometrically with mean 2 ($\Pr[h = j] = 2^{-j}$), giving expected per-odd-step drift

$$\delta(a) \;=\; \ln a - 2\ln 2 : \qquad \delta(1), \delta(3) < 0 < \delta(5) \le \delta(7) \le \cdots$$

The sweep measures halvings/odd steps $\in [1.9948,\ 2.0263]$ across all 38 variants, and exactly 2.0000–2.0009 on escape-dominated ensembles; measured drift matches $\delta(a)$ to *four decimals* for every $a \ge 5$ (Table 1). For the convergent $a = 3$ family the small deviations are exactly the finite-orbit boundary term

$$\frac{H_{\text{tot}}}{k_{\text{tot}}} \;=\; \log_2 3 \;+\; \frac{\overline{\log_2 n_{\text{start}}} - \overline{\log_2 n_{\text{end}}}}{\bar{k}}.$$

The law is also *kinetic*, not merely a sign: mean escape times obey $\bar{k} \approx (\log_2 \tau - \overline{\log_2 n_0}) \ln 2/\delta(a)$ — predicted ≈ 103 vs. measured 104.66 odd steps for $(5, 1)$, and ≈ 40 vs. 41.39 for $(7, 1)$.

Consequence. $a = 3$ is the unique odd multiplier > 1 with negative drift: the *only* member of the family that is both nontrivial and (heuristically) globally convergent. The family thus splits into a contracting regime $a \le 3$ and an expanding regime $a \ge 5$, and the data supports the split at 10^9–10^{10} starts per variant.

9. Synthesis: the status of the general case

For a, c odd, positive, $\gcd(a, c) = 1$ — Theorems 1–3 reduce everything else to this case:

claim	status
a or c even $\Rightarrow$ monotone divergence of every odd orbit	**proved** (Thm 1, Lean)
$a = 2^k - 1 \Rightarrow$ cycle through c, for every odd c	**proved** (Thm 4, Lean)
$F_{a,c}(c) = \mathrm{odd}(a{+}1)\,c$; fixed point at c **iff** $a = 2^k - 1$	**proved** (Thm 4b, Lean); zero violations on 2.75×10^{11} pairs
one-step cycles $\leftrightarrow$ divisors $d \mid c$ with $a + d \in 2^{\mathbb{N}}$	**proved** (Thm 4c, Lean); census of 1.37×10^{11} triples matches the law set-exactly

claim	status
$ax+1$ cycles transport to every $ax+c$ (universal families)	**proved** (Thm 4d, Lean); catalog: exactly 25 families through c for $a < 2^{24}$ (24 Mersennes + a=5)
scaling and absorption reductions	**proved** (Thms 2–3, Lean)
each cycle/inventory-completeness claim of §6	**machine-certified** (89 `decide` + 38 `native_decide`)
cycle structure ⇔ arithmetic of $2^H - a^k$	exact identity, verified on all 89 cycles
Conjecture A (generalized Collatz; Belaga–Mignotte): $a \le 3 \Rightarrow$ every orbit eventually periodic	**open**; supported at 100.0000% over $\approx 2.1 \times 10^9$ starts × 12 variants; "almost all" known for $(3, 1)$ (Tao)
Conjecture B (generalized Crandall): $a \ge 5 \Rightarrow$ divergent orbits exist and convergence density is 0	**open**; supported at $\approx 100\%$ escape over 26 variants; *no single orbit* (e.g. $n = 7$ under $5x + 1$) is provably divergent
uniform decision procedure for the family	**impossible** (Conway; Kurtz–Simon, Π^0_2-complete)

Table 3. What is proved, what is certified, what is open, what is impossible.

10. Why this route cannot close the conjecture — and what it did instead

Three separate walls, in increasing order of finality. **(1) Finite verification proves nothing asymptotic:** our 2^{32}, like Barina's 2^{71}, leaves the next integer unconstrained. **(2) The drift argument is probabilistic:** $\delta(3) < 0$ says orbits contract *on average over parity vectors*; the conjecture quantifies over *every* orbit, and no ergodic statement excludes a measure-zero parity sequence conspiring forever. The best unconditional result — Tao's "almost all orbits attain almost bounded values" — is precisely the strongest sentence this kind of argument seems able to produce; the gap "almost all → all" *is* the conjecture. **(3) Undecidability:** by Conway and Kurtz–Simon, no theorem schema can uniformly settle the generalized family; hardness is not an artifact of technique.

What survives these walls is exactly what a compute-plus-proof pipeline can deliver, and did: the provable general structure of the family (Theorems 1–4), certified at the kernel level; complete certified cycle inventories below explicit bounds for 38 systems; and two quantitative laws — drift $\delta(a) = \ln a - 2\ln 2$ and the signature mechanism $D = 2^H - a^k$ — that turn the family's phenomenology (why $3x+13$ has ten cycles, why $9x+1$ has none, why $a = 3$ alone converges) from folklore into measured, machine-checked fact. The Collatz conjecture, so embedded, stops looking like an isolated curiosity: it is the unique negative-drift member of a family whose cycles are governed by the Diophantine proximity of 2^H to a^k — and both of its remaining questions are transcendence-hard.

11. Reproducibility

```
nvcc -O3 -gencode arch=compute_90,code=compute_90 collatz.cu -o collatz_gpu
./collatz_gpu > results/raw.jsonl        # ~10 s on an RTX 5090
python3 analyze.py                       # exact rechecks + asserts + generates CollatzCerts.lean
lean CollatzTheory.lean                  # general theorems incl. §4.1 (Lean 4.31.0, no dependencies)
lean CollatzCerts.lean                   # 127 certificates (~6 min, native_decide)

nvcc -O3 -gencode arch=compute_90,code=compute_90 universal.cu -o universal_gpu
./universal_gpu > results/universal.jsonl   # S1-S4 theorem verification, ~2 s
python3 verify_universal.py                 # exact-arithmetic layer V1-V8 + catalog summary
```

Hardware: NVIDIA RTX 5090 (32 GB), driver 595.71.05 (CUDA 13.2), `nvcc` 12.0 via PTX JIT. All raw output, logs, and the generated certificate file are committed alongside this paper.

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